COS 397: Computer Science Capstone I

System Requirements Specification Template

(Adapted from Susan Mitchell and Michael Grasso)

General Instructions

1. Provide a cover page that includes the document name, product name, customer name, team name, team member names, and the current date. Include a team logo if you have one. You should include a version number for this document.
2. Number the pages of the document. Include page numbers in the table of contents.
3. Number and label all figures. Each figure or table should have a number, title (brief descriptive name of the figure), and a caption (a longer description of what the reader is supposed to understand from the figure or table). Refer to the figures and tables by number in the text.
4. All sections should have an introductory sentence or two.
5. Do not use vague words and phrases such as may, might, could, possibly, assumed to be, some, a little, and a lot. Use strong, definite words and phrases such as shall, will, will not, can, and cannot. Words such as "should" can be used to show suggestions, but use it sparingly.
6. Watch your spelling, punctuation, and grammar. It is a reflection on your professionalism.

Be sure that your document is

- Complete – No information is missing
- Clear – Every sentence's meaning must be clear to all parties
- Consistent – The writing style and notation is consistent throughout the document and the document does not contradict itself
- Verifiable – All facts stated are verifiable

Remember that you are required to do a peer review of this document.

When you think you are done with the SRS, ask yourself, "Could someone who was not part of the development of this SRS write the corresponding System Design Document?"

[Put team logo here]

[Put product name here]

Software Requirement Specification

Enabling 3D Printing of a Medical CT-Scan:

A Web App for Patients and Practitioners

Prepared by The Slice Is Right

COS 397 - Computer Science Capstone 1

October 20th, 2025

Version 1.0.0

System Requirements Specification

Table of Contents

1. Introduction

This capstone project, completed as part of the requirements for the Bachelor of Science in Computer Science at the University of Maine, focuses on developing a web application called *“Enabling 3D Printing of a Medical CT-Scan: A Web App for Patients and Practitioners.”* The main goal of this project is to create a browser-based tool that gives users the ability to upload medical CT scans in the DICOM format and turn them into files that can be used for 3D printing. The application is designed to produce both polygonal (.stl) files, which can be used to visualize different anatomical structures like bone, skin, or muscle, and G-code files, which are needed for 3D printers to accurately reproduce tissue characteristics. One of the most important aspects of this project is that all processing is done locally on the user's computer, which helps protect sensitive medical data and makes the tool more accessible for both medical research and patient education. By working on this project, I am contributing to the University of Maine’s ongoing efforts to advance medical imaging and 3D printing, especially through the Laboratory for Convergent Science.

1.1 Purpose of This Document

This document is meant to lay out the scope, design goals, and reasoning behind the proposed capstone project. It acts as a starting point for planning and requirements, giving the development team, faculty supervisors, and project sponsor a clear idea of what is expected. Here, the system’s purpose, background, and context are explained so that everyone involved understands what the final product should accomplish and how it will be judged. The intended audience includes Dr. Terry Yoo, the University of Maine Computer Science faculty, and any future students or researchers who might build on this work. The document goes on to provide background information and references, sets out the main objectives for the system, and explains where the system ends and outside elements like users, input files, and output devices begin.

1.2. References

Primary Project Source

Yoo, Terry. *Enabling 3D Printing of a Medical CT-Scan: A Web App for Patients and Practitioners.* University of Maine, Laboratory for Convergent Science, 2025.

Web and Technical References

- “DICOM Standard.” National Electrical Manufacturers Association (NEMA). Available at: https://www.dicomstandard.org/
- “Marching Cubes Algorithm for 3D Surface Construction.” Wikipedia. Available at: https://en.wikipedia.org/wiki/Marching_cubes
- “STL (File Format) Specification.” 3D Systems. Available at: https://www.3dsystems.com/support
- “G-code Reference.” RepRap Wiki. Available at: https://reprap.org/wiki/G-code websites).

1.3. Purpose of the Product

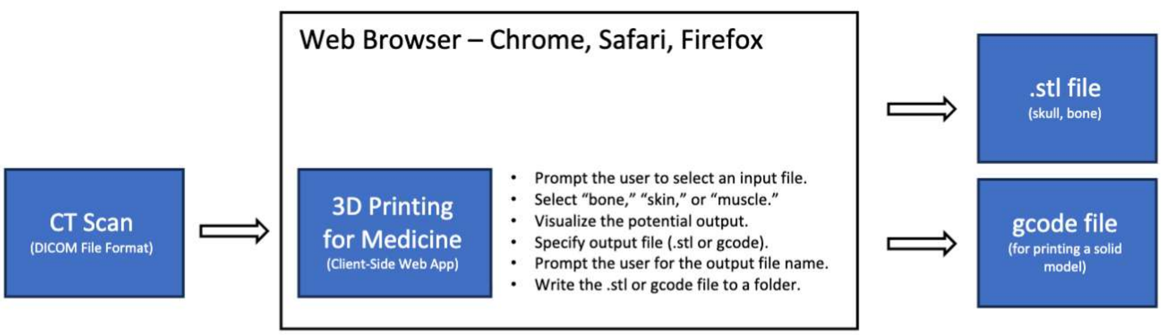
Dr. Terry Yoo, who serves as the Director of the Laboratory for Convergent Science at the University of Maine, has dedicated his research to the fields of medical imaging and 3D printing. His primary focus is on taking complex medical data and turning it into physical models that can be used for both educational and research purposes. The main motivation for this project comes from a clear need within the medical community: there is currently no secure, browser-based tool that enables patients and practitioners to easily visualize and 3D print anatomical structures directly from CT scans. Most existing solutions require third-party software or cloud-based processing, which can introduce privacy concerns and complicate the workflow. To address this, the goal is to develop a functional, open-source web application that can convert DICOM files into both STL and G-code formats. This would make it much easier to prepare CT scan data for 3D printing, while also making these tools more accessible to researchers and educators working in medical imaging.

1.4. Product Scope

This project focuses on creating a web application that gives users the ability to turn CT scan data into files ready for 3D printing, all from within their own browser. The main goal is to make the process straightforward and accessible, so that users can handle everything themselves without needing extra software or sending their data anywhere else. The system will allow users to:

- Upload a medical CT scan in the DICOM format.
- Choose an anatomical layer (bone, skin, or muscle).
- Visualize the generated 3D model.
- Select the desired output format (.stl or G-code).
- Download the resulting file locally for use in 3D printing.

Every part of the process happens right in the user's web browser, whether that is Chrome, Safari, or Firefox. This means that all medical data stays on the user's own device at all times. There is no use of cloud storage, outside databases, or any network-based processing. By keeping everything on the client side, the system protects user privacy and keeps things as simple as possible.



2. Functional Requirements

It is recommended that each functional requirement should be represented using a use case.

Refer the reader to the top-level use case/context diagram referred to in Section 1.4. In addition, include separate use case diagrams, where appropriate, for each of the top-level use cases.

Number	< use case number >	
Name	< use case name - a short active verb phrase >	
Summary	< a brief summary of the use case >	
Priority	< how critical this use case is to the customer (1 to 5, 5 being most critical >	
Preconditions	< conditions that must be true before the use case trigger >	
Postconditions	< conditions that will be true after the use case completes >	
Primary Actor	< a role name for the primary actor >	
Secondary Actors	< other systems that are relied upon to accomplish the use case >	
Trigger	< the action that starts the use case >	
Main Scenario	**Step**	**Action**
	1	< steps of the use case from trigger to goal delivery >
	2	< … >

	3	< … >
Extensions	**Step**	**Branching Action**
	1a	< condition causing branching > **:** < action or name of sub use case >
Open Issues	< list of issues awaiting decisions that affect the use case >	

In addition to the diagrams, every use case should be documented using the following use case specification format.(This template was adapted from Alistair Cockburn.)

Lastly, write the tests that will be used during system and acceptance testing to verify that each requirement has been met. Note that a single requirement may require multiple tests, so be thorough. It is also possible that a single test verifies more than one requirement. The goal is to come up with the minimum number of test cases that thoroughly test the system. Make sure that the test numbers correspond to the use case numbers.

Functional Requirements:

1. DICOM File Import
    a. The System shall allow the user to select and upload a DICOM file(or folder of DICOM files) from their local machine
    b. The system shall validate the DICOM format before processing
    c. The system shall read and parse the DICOM file to extract relevant CT scan image data and metadata(or - information?)
2. Anatomical Region Selection
    a. The system shall allow the user to select an anatomical target for visualization and output - bone, skin, muscle
    b. The system shall filter or threshold CT data based on the selected tissue type
    c. The system shall update any visualization or preview accordingly when a different region is selected - going from bone to muscle updates automatically
3. 3D Visualization
    a. The system shall generate and display a 3d preview of the reconstructed anatomy from the CT data within the web browser
    b. The visualization shall allow the user to rotate, zoom, and inspect the 3D model interactively

   c. The visualization shall update dynamically based on user selection(tissue type or output format)
4. Polygonal Model Generation(STL conversion)
   a. The system shall apply a Marching Cubes (or similar) algorithm to extract a polygonal surface representation of the selected anatomy
   b. The system shall generate a 3D mesh from the processed CT scan data
   c. The system shall export the resulting 3D model as a .STL file suitable for 3D printing or slicing
5. G-Code Generation
   a. The system shall allow the user to select an output type: .stl or .gcode
   b. The system shall generate G-code output that reproduces the Xray attenuation properties of the tissue - mimicking density or opacity
   c. The system shall write the generated G-code to a user-specified output location on the local machine
6. File Output
   a. The system shall prompt the user to enter or confirm an output file name
   b. The system shall save the resulting .stl or .gcode to a local folder chosen by the user
   c. The system shall notify the user when the file generation is complete or if there is an error
7. User Interface
   a. The system shall provide an intuitive and responsive graphical user interface that allows users to:
      i. Select input files(DICOM)
      ii. Choose anatomy type(bone, skin, muscle)
      iii. Choose output type(.stl or .gcode)
      iv. Visualize results in 3D view
      v. Choose output file name and location
8. Client-Side Execution
   a. The system shall run entirely within the client's browser with no external data transfer
   b. The system shall process DICOM data locally, ensuring that medical data never leaves the user's machine
9. Cross-Browser and Cross-Platform Compatibility
   a. The system shall operate correctly on Chrome, Safari, and Firefox
   b. The system shall function across MacOS, Windows, and Linux operating systems
10. Automated Testing and Verification
   a. The system shall include an automated test suite to verify correct operation across supported browsers and OS platforms
   b. The system shall use a web testing framework (e.g., Selenium) to validate core functionality

Use Case Specifications:

Use Case -1: Import DICOM

Number	1	
Name	Import DICOM File	
Summary	The user selects and uploads a DICOM file or folder of DICOM files containing CT scan data for processing	
Priority	5	
Preconditions	1. The web app is loaded in the user's browser 2. The user has valid DICOM data stored locally	
Postconditions	1. The DICOM file is validated and parsed 2. CT image data and metadata are ready for processing	
Primary Actor	User	
Secondary Actors	Web browser file Input/Output API	
Trigger	User selects "Upload DICOM" or equivalent option in the interface	
Main Scenario	**Step**	**Action**
	1	The user opens the web app
	2	The system prompts the user to select a DICOM file or folder
	3	The user selects one or more DICOM(.dcm) files from their local machine

	4	The system validates the selected files to make sure they're in the DICOM format
	5	The system parses the DICOM data and extracts image slices and metadata
	6	The system notifies the user that the import was successful or reports an error if validation fails
Extensions	**Step**	**Branching Action**
	4a	If the selected file is not a valid DICOM format **:** The system displays an error and prompts the user to try again
	5a	If the file is corrupted: The system alerts the user and terminates the import process
Open Issues	None	
Verification Tests	1. Load valid DICOM: Try with different sizes and number of files. 2. Invalid file: error shown	

Use Case -2: Select Anatomical Region

Number	2
Name	Select Anatomical Region
Summary	The user chooses which anatomical region(bone, skin, or muscle) to visualize and output

Priority	4	
Preconditions	User case 1 complete	
Postconditions	Segmentation parameters for the target are active.	
Primary Actor	End user	
Secondary Actors	● DICOM Processing Module ● Browser Interface	
Trigger	User selects tissue option	
Main Scenario	**Step**	**Action**
	1	The system displays tissue options
	2	User selects one.
	3	System applies preset thresholds/range
Extensions	**Step**	**Branching Action**
	1a	Custom thresholds parameterized by user
Open Issues	None	
Verification Tests	1. Selecting “bone” applies bone preset 2. Try all the other presets	

Use Case -3: Preview Output

Number	3	
Name	Visualize and Inspect 3D Anatomy	
Summary	App previews the selected anatomy, like surface or slices	
Priority	4	
Preconditions	User cases 1 and 2 completed	
Postconditions	Preview is rendered user can accept or adjust	
Primary Actor	End user	
Secondary Actors	• Rendering Engine: WebGL/WebGPU - generates real-time previews • Segmentation Engine - provides voxel data for rendering • UI Controller - manages preview controls	
Trigger	User opens Preview	
Main Scenario	**Step**	**Action**
	1	System generates isosurface/volume preview
	2	User rotates and zooms
	3	toggles between different levels of threshold
	4	User accepts preview
Extensions	**Step**	**Branching Action**

	1a	Takes a long time or a large input file**:** Lower resolution preview
Open Issues	None	
Verification Tests	1. Preview appears within N seconds on each supported browser 2. Threshold change updates preview	

Use Case -4: Generate STL

Number	4
Name	Create Polygonal Mesh (.stl export)
Summary	Produce an STL mesh of the selected anatomy via Marching Cubes and exports it as an .stl file
Priority	5
Preconditions	User case 1 and 2 completed
Postconditions	STL mesh is created in memory and an .stl file gets downloaded
Primary Actor	End User
Secondary Actors	● Marching Cubes Algorithm Module: performs isosurface extraction ● Mesh Post-Processing Engine: repairs or smooths geometry ● System Memory: temporarily holds output mesh data
Trigger	User clicks “Generate STL”

Main Scenario	Step	Action
	1	System runs Marching Cubes with current thresholds.
	2	System post-processes - smoothing, missing spots
	3	Mesh statistics displayed
Open Issues	None	
Verification Tests	1. Output .stl is readable by standard slicers 2. Mesh bounds match CT bounds within tolerance	

Use Case -5: Generate G-code

Number	5
Name	Generate 3D Print Instructions(.gcode export)
Summary	Produce G-code that mimics tissue x-ray characteristics when printed
Priority	4
Preconditions	User case 1 and 2 completed
Postconditions	G-code held in memory and .gcode gets downloaded
Primary Actor	End User
Secondary Actors	Depends on the algorithm we get from our client

Trigger	User clicks “Generate G-code”	
Main Scenario	**Step**	**Action**
	1	Depends on the algorithm we get from our client
Extensions	**Step**	**Branching Action**
	1a	Optional hardware profile templates like: printer nozzle, materials
Open Issues	None	
Verification Tests	1. G-code loads in a standard host 2. Same output across all browsers	

3. Non-Functional Requirements

NFR-01 (Performance) Priority-5

Description - The system shall process and convert a DICOM dataset (≤ 500 MB) to an STL or G-code file within 120 seconds on a modern desktop machine.

Verification - Perform times conversion runs on benchmark DICOM files; confirm the output was within 120 seconds.

NFR-02 (Usability) Priority-4

Description - The user interface shall enable full operation (file import -> export) with no more than 5 major user interactions from the main screen.

Verification - Conduct usability testing with 5 users; record the number of clicks and ease-of-use ratings.

NFR-03 (Reliability) Priority-5

Description - The system shall process at least 95% of valid DICOM files without crashes, freezes, or corrupted outputs.

Verification - Run 20 test cases of varied DICOM inputs; verify successful, error-free operation in ≥ 95% of runs.

NFR-04 (Compatibility) Priority-4

Description - The system shall function identically across Chrome, Firefox, and Safari browsers on MacOS, Windows, and Linux.

Verification - Executed automated cross-browser tests; confirm consistent visualization and output results.

NFR-05 (Security & Privacy) Priority-5

Description - All computation shall occur locally within the user's browser; no external servers or data transfers are permitted.

Verification - Monitor that there is no outbound data when running the program during processing.

NFR-06 (Maintainability) Priority-3

Description - The codebase shall be modular, documented and maintain ≥ 80% functional-level comments for future modifications.

Verification - Perform static code analysis to assess documentation coverage and modular design compliance.

NFR-07 (Accuracy) Priority-5

Description - The generated STL and G-code outputs shall represent anatomical geometry within a ± 2% deviation of the source CT data.

Verification - Compare polygonal model geometry to CT voxel data using mesh validation tools.

NFR-08 (Scalability) Priority-2

Description - The application shall remain responsive when handling CT datasets up to 1 GB without major lag or browser crashes.

Verification - Perform stress testing with large datasets; measure system responsiveness and memory stability.

NFR-09 (Resource Efficiency) Priority-4

Description - The system shall maintain CPU utilization below 80% and memory usage below 2GB during DICOM file conversion and visualization on a modern desktop system.

Verification - Use browser performance profiling tools to measure CPU and memory usage during test conversions.

NFR-10 (Robustness) Priority-5

Description - The system shall handle malformed or incomplete DICOM data gracefully, displaying a clear error message without crashing or lost data.

Verification - Feed 20 intentionally corrupted or incomplete DICOM files; confirm no system crashes and proper user-facing error reporting.

4. User Interface

See “User Interface Design Document for Enabling 3D Printing of a Medical CT-Scan.” here.

[Will be linked when document is created]

5. Deliverables

Provide a list of all deliverable items (that is, all artifacts that you will deliver to the customer). This list will include items such as the product itself (What format? Source code? Executable code? Object code?), documentation, and training resources (if any). Specify when (date) and in what format (e.g., hard copy, zip file (how delivered, Git?)) each will be delivered. A tabular format works well for this section. We will assume that the deliverable items are as follows:

Hard copies of each of the following: ***[Will be linked when document is created]***

- Systems Requirement Specification
- System Design Document
- User Interface Design Document
- User Manual
- Administrator Manual
- Copies of all Biweekly Status Reports

An electronic file containing the following:

- Systems Requirement Specification

- System Design Document
- User Interface Design Document
- User Manual
- Administrator Manual
- All source code
- The executable program
- Any other software required for installation and execution of the delivered program.

6. Open Issues

Issues that have been raised and do not yet have a conclusion. These issues will be addressed later in the development process.

Appendix A – Agreement Between Customer and Contractor

This Software Requirements Specification document constitutes a formal agreement between the Customer, Terry Yoo, and the Contractor, The Slice Is Right, regarding the functional and non-functional requirements for the Enabling 3D Printing of a Medical CT-Scan system. By signing below, both parties acknowledge that this document accurately and completely captures the mutual understanding of the system to be developed. The Customer agrees that this SRS provides a sufficient basis for the Contractor to proceed with the system design and implementation, and the Contractor agrees to develop a system that conforms to the requirements described above.

Any future changes, additions, or modifications to the requirements specified in this document must be managed through a formal change control process. A request for change must be submitted in writing by either party and will be evaluated for its impact on project scope, schedule, and feasibility. An amended version of this SRS, or a formal change order referencing this document, must be mutually agreed upon and signed by authorized representatives of both the Customer and the Contractor before any changes are implemented in the project.

When Signing:

In the “Name (Typed)” section, please replace the *[TYPE NAME HERE]* text with your name.

In the “Signature” section, click the “Line” icon in the ribbon, , then select “Scribble”, and sign in the slot allowed.

In the “Date” section, please replace the *[TYPE DATE HERE]* text with the date signed.

Make sure to “save and close” in the top right corner to finish signing the document.

Contractor Sign-off

We, the undersigned, hereby agree that the requirements described in this document are understood and will be met by the delivered system.

Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]
Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]
Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]
Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]
Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]

Customer Sign-off

We, the undersigned, hereby approve the requirements described in this document and authorize the Contractor to proceed with the project.

Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]

Customer Comments

[Please provide any additional comments, assumptions, or acknowledgments here.]

Appendix B – Team Review Sign-off

This document confirms that all undersigned members of the The Slice Is Right project team have thoroughly reviewed the entirety of this Software Requirements Specification for the Enabling 3D Printing of a Medical CT-Scan system. By signing below, each team member acknowledges their understanding of the requirements and formally agrees that the content, scope, and structure of this document are accurate and complete, providing a suitable foundation for the subsequent phases of the project.

The comment section provided for each team member is to be used for noting any minor suggestions, editorial feedback, or non-substantive points of clarification. It is recognized that for this sign-off to be granted, there are no major points of contention regarding the technical or functional requirements outlined within this SRS.

Contractor Sign-off

We, the undersigned, hereby agree that the requirements described in this document are understood and will be met by the delivered system.

Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]
Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]
Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]
Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]
Name (Typed)	[TYPE NAME HERE]	Signature		Date	[TYPE DATE HERE]

Comments

[Please provide comments if there are any minor points that members may not agree with]

Appendix C – Document Contributions

This appendix details the contributions of each team member to the creation of this Software Requirements Specification. All members participated in the collaborative writing, review, and diagramming process to ensure a comprehensive and unified document. The percentage contributions are estimates that reflect the primary authorship and development effort for the various sections.

Member Name	Primary Responsibilities	Estimated Percentage
Israk Arafat	Functional Requirements	20%
Gregory Michaud	Introduction Section	20%
Cooper Stepankiw	Title Cover Team Logo Section 4,5 Appendix A,B,C	20%
Bryan Sturdivant	Functional Requirements	20%
Ethan Wyman	Section 3; Non-Functional Requirements Table of Contents Formatting	20%
Total		100%

Name (Typed) [TYPE NAME HERE] Signature Date [TYPE DATE HERE]

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